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Averaging Methods:

Boosting Methods → Train one algo
↳ 3 subtypes → errors of one, fix the next one.

→ Arrange in sequence

→ errors of one → update the next one.

Stacking → Different models prediction (Base mode

In last apply another one.

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Boosting **Types of Ensemble Learning**

↳① Subtype . Bagging Algorithm

② . Gradient Boosting

③ . AdaBoost

④ . XG Boost

⑤ . Stacking Algorithm

⇒ Gradient Boosting

⇒ Adaboost

⊙ First model → wrong prediction

→ Its focus on First tree → model

↳ not overall.

⇒ XG Boost

Gradient Boost → Advanced version

⇒ Stacking

⊙ Three models → individual prediction

⊙ Then apply one in last to make final prediction → for validation.

VOTING CLASSIFIER

⊙ Hard	⊙ Soft
Each model predicts a class label	Average the predicted class probabilities

voting = 'hard' or voting = 'soft'

bootstrap = `True´ = `False´

↳ we cannot assign this value to anyone.

Gradient

no. of estimators = ?

learning rate = ?

Stacking

3 Base models

↓ ↓ ↘

Random Forest | Logistic Regression | Decision Tree.

Make the final one

Ada boost

which classifier = ?

depth : ?

class_weight = balanced.

no. of estimators = ?

learning rate = ?